Freelance Platform

Complete Project Documentation

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A modern freelancing marketplace built with Angular/Ionic and Flask, connecting clients and skilled freelancers through an intuitive platform.

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1. Project Overview

Freelance Platform is a modern, full-stack web and mobile application that connects freelancers with clients. It provides a seamless marketplace where clients can post projects and freelancers can submit competitive proposals.

Built with **Angular 20 / Ionic 8** on the frontend and **Flask** on the backend, with **MongoDB** as the database, the platform offers a responsive cross-platform experience accessible from web browsers and mobile devices (iOS and Android via Capacitor).

The project aims to replicate the core functionality of established platforms such as Upwork or Freelancer.com, tailored for academic and professional demonstration purposes.

2. Objectives

- Connect clients and freelancers through a structured project marketplace.
- Streamline project management with milestones, budgets, and deadlines.
- Provide secure authentication using JSON Web Tokens (JWT).
- Enable proposal management allowing freelancers to bid on projects.
- Support messaging between clients and freelancers.
- Offer subscription tiers for premium features and visibility.
- Deliver a mobile-first experience through Ionic/Capacitor for iOS and Android.
- Demonstrate full-stack integration between a modern frontend framework and a Python REST API.

3. Features and Functionalities

3.1 Implemented Features

Backend (Flask REST API)

✓ 14 REST Endpoints — fully documented and tested
✓ Project CRUD — Create, Read, Update, Delete projects
✓ Proposal CRUD — Submit, view, update, delete proposals
✓ JWT Authentication — Token-based endpoint protection
✓ Search & Filtering — Full-text search and filter by status/category/skill
✓ Error Handling — Structured JSON error responses
✓ CORS Configuration — Cross-origin resource sharing for frontend integration
✓ MongoDB Integration — Persistent data storage with PyMongo

Frontend (Angular / Ionic)

✓ Onboarding Screen — Introduction flow for new users
✓ Authentication Pages — Login and Sign-up screens
✓ Home Dashboard — Featured freelancers, categories, quick navigation
✓ Projects Listing — Browse, search, and filter available projects
✓ Project Details — Full project information with proposal submissions

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✓ Create Project — Form for clients to post new projects
✓ Create Proposal — Freelancers submit bids with cover letters
✓ Profile Management — View and edit user profile information
✓ CV Management — Upload and manage curriculum vitae
✓ Messaging — Direct messaging (message list, recipients, chat screen)
✓ Pricing / Subscription — Plan selection and subscription success flow
✓ JWT Interceptor — Automatic token injection in HTTP headers
✓ Loading States — Spinners and skeleton loaders for async operations
✓ TypeScript Strict Mode — Fully typed codebase

3.2 In-Progress Features

■ Full authentication flow (login/register/logout/refresh)
■ Real-time notifications via WebSockets
■ Rating and review system
■ Integrated payments

3.3 Planned Features

■ Admin panel and analytics dashboard
■ Multi-language support (i18n)
■ Offline mode and Progressive Web App (PWA)
■ Native mobile builds (iOS / Android)

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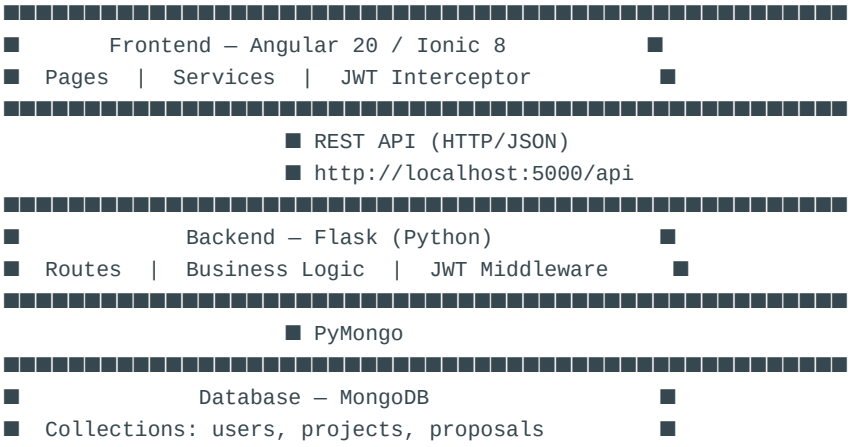
4. System Architecture

4.1 High-Level Architecture

The platform follows a classic three-tier client-server architecture:

Tier	Technology	Role
Presentation	Angular 20 / Ionic 8	User interface, mobile/web client
Application	Flask (Python)	Business logic, REST API, JWT auth
Data	MongoDB	Document storage, persistence



■■■

5. Technologies Used

5.1 Frontend

Technology	Version	Purpose
Angular	20.0	Web application framework
Ionic Framework	8.0	Mobile-first UI components
Capacitor	8.0	Native mobile runtime (iOS/Android)
TypeScript	5.9	Statically typed JavaScript
RxJS	7.8	Reactive programming (Observables)
Angular HttpClient	20.0	HTTP communication
Ionicons	7.0	Icon library

5.2 Backend

Technology	Version	Purpose
Python	3.8+	Backend runtime
Flask	2.3.3	Lightweight web framework
PyMongo	4.5.0	MongoDB driver for Python
PyJWT	2.11.0	JWT generation and validation
Flask-CORS	4.0.0	Cross-Origin Resource Sharing

5.3 Database

Technology	Version	Purpose
MongoDB	6.0+	NoSQL document database

5.4 Development Tools

Tool	Purpose
Angular CLI	Code generation, build, and serve
Ionic CLI	Ionic project management and preview
npm	Node.js package manager
pip	Python package manager
Git	Version control
VS Code	Recommended IDE
Postman	API testing

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6. Installation and Setup

6.1 Prerequisites

Requirement	Version	Download
Python	3.8+	https://www.python.org
Node.js	16+	https://nodejs.org
MongoDB	6.0+	https://www.mongodb.com
Git	Any	https://git-scm.com

6.2 Clone the Repository

```
git clone https://github.com/sellami-mohammed-aziz/Freelance-Plateform.git
cd Freelance-Plateform
```

6.3 Backend Setup

```
cd backend

# Create and activate virtual environment
python -m venv venv
source venv/bin/activate       # Linux/Mac
# venv\Scripts\activate       # Windows

# Install dependencies
pip install -r requirements.txt

# Configure environment variables
cp .env.example .env
# Edit .env:
#   MONGO_URI=mongodb://localhost:27017/freelance_platform
#   JWT_SECRET_KEY=your-secure-secret-key
```

6.4 Frontend Setup

```
# From project root
npm install -g @ionic/cli
npm install
```

6.5 Database Setup

```
# Start MongoDB service
mongod                         # Linux/Mac
net start MongoDB              # Windows (as service)
```

MongoDB automatically creates the freelance_platform database and collections on first use.

6.6 Running the Application

Option A — Automatic (Recommended):

```
./run.sh       # Linux/Mac
.\run.bat      # Windows
```

Option B — Manual:

```
# Terminal 1 — Backend
cd backend && python run.py
# → http://localhost:5000/api

# Terminal 2 — Frontend
ionic serve
# → http://localhost:4200
```

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7. Usage Guide

7.1 First Launch

1. Open your browser and navigate to http://localhost:4200
2. The app launches on the Onboarding screen
3. Proceed to Sign Up to create a new account, or Login if you already have one

7.2 Client Workflow

1. Login with client credentials
2. Navigate to Home dashboard
3. Go to Projects → click New Project (/projects/new)
4. Fill in: title, description, budget, deadline, skills required
5. Submit — the project is published and visible to freelancers
6. Review incoming Proposals from your project details page
7. Accept the best proposal to begin work

7.3 Freelancer Workflow

1. Login with freelancer credentials
2. Browse Projects at /projects
3. Use the Search bar or Category filters to find relevant projects
4. Open a Project Details page to read full requirements
5. Click Submit Proposal: enter bid amount, delivery time, cover letter
6. Track proposal status from your profile

7.4 Messaging

1. Navigate to the Messages section
2. Select a recipient from Message Recipients
3. Send and receive messages in the Messaging Screen

7.5 Profile Management

1. Navigate to Profile page
2. Click Edit Profile to update name, bio, skills, and contact info
3. Use CV Management to upload your resume (PDF/DOC)

7.6 Subscription

1. Navigate to Pricing (/upgrade)
2. Review available plans (Free, Pro, Enterprise)
3. Select a plan and complete checkout
4. View Subscription Success confirmation

8. API Endpoints

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Base URL: **http://localhost:5000/api** | Authentication: Include *Authorization: Bearer <jwt_token>* header for protected routes.

8.1 Projects

Method	Endpoint	Auth	Description
GET	/projects	No	List all projects (pagination, filtering)
POST	/projects	JWT	Create a new project
GET	/projects/	No	Get project details by ID
PUT	/projects/	JWT	Update a project
DELETE	/projects/	JWT	Delete a project
PATCH	/projects//status	JWT	Update project status
GET	/projects/search?q=...	No	Search projects by keyword
GET	/projects/client/	No	Get projects by client ID

Example — Create Project Request:

```
POST /api/projects
Authorization: Bearer <token>
Content-Type: application/json

{
  "title": "Build an e-commerce website",
  "client": "John Doe",
  "budget": 2500,
  "deadline": "2026-06-30",
  "category": "Development",
  "skills": ["React", "Node.js", "MongoDB"]
}
```

8.2 Proposals

Method	Endpoint	Auth	Description
GET	/proposals/	No	List proposals for a project
POST	/proposals	JWT	Submit a new proposal
GET	/proposals/	No	Get proposal details by ID
PUT	/proposals/	JWT	Update a proposal
DELETE	/proposals/	JWT	Withdraw/delete a proposal
PATCH	/proposals//status	JWT	Update proposal status

8.3 Authentication (Planned)

Method	Endpoint	Description
POST	/auth/register	Register a new user
POST	/auth/login	Login and receive JWT token

Method	Endpoint	Description
POST	/auth/logout	Invalidate current token
POST	/auth/refresh-token	Obtain a refreshed token

8.4 Standard Error Responses

```
// 400 Bad Request
{ "error": "Missing required field: title" }

// 401 Unauthorized
{ "error": "Token manquant" }

// 404 Not Found
{ "error": "Project not found" }

// 500 Internal Server Error
{ "error": "Internal server error" }
```

9. UI Descriptions

9.1 Onboarding Screen

An introductory carousel presenting the platform's value proposition. Users can skip or proceed to Sign Up / Login. Features animated illustrations and prominent call-to-action buttons.

9.2 Home Dashboard

Header with user greeting, location badge, and notification bell. Horizontal scrollable category bar (All, Design, Development, Writing, Marketing, Video, Music). Featured freelancer cards with avatar, name, specialty, and rating. Guest banner for unauthenticated users. Persistent bottom navigation tabs.

9.3 Projects Screen

Full-text search bar across titles and descriptions. Quick-filter chips by status (Pending, In Progress, Completed). Project cards showing title, client avatar, budget, deadline, category badge, and proposal count. Floating action button for clients to create new projects.

9.4 Project Details Screen

Header with title, client info, and status badge. Formatted description paragraphs. Skill tag badges. Budget and deadline info cards. Milestone timeline with status indicators. Downloadable attachment list. Proposal list and submit-proposal action button.

9.5 Create Project Screen

Form fields: Title, Category (dropdown), Budget, Deadline (date picker), Duration, Description (multi-line), Required Skills (tag input), dynamic Milestones list, and file Attachments upload.

9.6 Profile Screen

Avatar and cover image. Name, role, bio. Stats: projects completed, rating, review count. Skill badges. Action buttons: Edit Profile, Manage CV, View Proposals.

9.7 Messaging Screen

Message list with conversation threads, last message preview, and timestamp. Chat screen with message bubbles and text input bar. Recipients screen with searchable contact list to start new conversations.

9.8 Pricing Screen

Three-tier pricing cards: Free (basic access, limited proposals), Pro (unlimited proposals, featured listing, priority support), Enterprise (custom contracts, dedicated account manager). Subscription success confirmation screen after checkout.

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10. Future Improvements

10.1 Short-Term (Next Quarter)

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Feature	Priority	Description
Complete Authentication	HIGH	Full login/register/logout/refresh token flow
User Roles	HIGH	Distinct permissions for Client vs. Freelancer
Public Profiles	HIGH	Public freelancer profile with portfolio
Notifications	MEDIUM	In-app and push notifications
Rating System	MEDIUM	5-star ratings and reviews after project completion

10.2 Medium-Term

Feature	Priority	Description
Payment Integration	HIGH	Stripe or PayPal for milestone payments
Real-Time Messaging	MEDIUM	WebSocket-powered live chat
Advanced Search	MEDIUM	Elasticsearch full-text and semantic search
Analytics Dashboard	MEDIUM	Client and freelancer activity dashboards
Admin Panel	MEDIUM	Platform moderation and user management

10.3 Long-Term

Feature	Priority	Description
Native Mobile Apps	MEDIUM	iOS and Android builds via Capacitor
Progressive Web App	MEDIUM	Offline support and installable web app
Multi-Language	LOW	i18n for Arabic, French, and English
AI Matching	LOW	ML-based freelancer-to-project matching
Video Interviews	LOW	Integrated video calls for client/freelancer meetings
Escrow System	LOW	Secure milestone-based fund release

10.4 Technical Improvements

- Testing: Unit tests for all services, E2E tests with Cypress
- CI/CD Pipeline: GitHub Actions for automated build and deployment
- Docker Containerization: Docker Compose for local and production environments
- Performance: Redis caching, CDN for assets, lazy-loading optimization
- Security: Rate limiting, input sanitization, HTTPS enforcement, 2FA
- Monitoring: Application logging with ELK Stack or Sentry

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Appendix A — Project File Structure

```
Freelance-Plateform/
■■■ docs/
■   ■■■ Project_Documentation.md    ← Source documentation
■   ■■■ Project_Documentation.pdf   ← Generated PDF (this file)
■
■■■ src/                            ← Angular/Ionic frontend
■   ■■■ app/
■   ■   ■■■ pages/                   ← 16 application screens
■   ■   ■■■ components/              ← Reusable UI components
■   ■   ■■■ core/services/           ← 7 service modules
■   ■   ■■■ models/                  ← 7 TypeScript interfaces
■   ■   ■■■ interceptors/            ← JWT interceptor
■   ■   ■■■ app.routes.ts            ← Route definitions
■   ■■■ environments/
■
■■■ backend/                        ← Flask REST API
■   ■■■ app/
■   ■   ■■■ app.py
■   ■   ■■■ config.py
■   ■   ■■■ models.py
■   ■   ■■■ routes.py
■   ■■■ run.py
■
■■■ README.md
■■■ angular.json
■■■ package.json
■■■ ionic.config.json
```

Appendix B — Environment Variables

```
# backend/.env
FLASK_ENV=development
FLASK_DEBUG=1
MONGO_URI=mongodb://localhost:27017/freelance_platform
JWT_SECRET_KEY=your-secure-random-secret-key
JWT_EXPIRATION_HOURS=24

# src/environments/environment.ts
export const environment = {
  production: false,
  apiUrl: 'http://localhost:5000/api'
};
```

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